Olympiad Test Paper — Science (Class 7)

Chapter 18: Wastewater Story

Subject	Science (NCERT Class 7)
Chapter	18 — Wastewater Story
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. All questions are answerable from the Class 7 chapter "Wastewater Story" — no outside facts are needed. Many questions test the *order* and *purpose* of treatment steps, the difference between aerobic and anaerobic stages, and common traps (treated water is cleaner but not necessarily drinkable). Read each stem carefully. Do not write on this paper — mark answers on your answer sheet.

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1. Used water released from homes, industries and farms is collectively given the name:

(A) sludge
(B) scum
(C) sewage
(D) biogas

2. Sewage is best described as:

(A) a solid waste left behind after water dries up
(B) a liquid carrying suspended and dissolved impurities
(C) a gas given off by rotting waste
(D) pure rainwater collected from rooftops

3. The network of underground pipes that carries sewage away from a city is called the:

(A) sewerage
(B) aeration tank
(C) digester
(D) grit chamber

4. Which of these is **not** normally a part of household wastewater?

(A) water from washing clothes
(B) water from the kitchen sink
(C) molten iron from a furnace
(D) water flushed from the toilet

5. The impurities present in sewage are together referred to as:

(A) nutrients only
(B) contaminants
(C) minerals
(D) sediments only

6. Human faeces, leftover food and cooking oils in sewage are examples of:

(A) inorganic impurities
(B) dissolved metals
(C) harmless salts
(D) organic matter

7. Nitrates and phosphates dissolved in sewage are classified as:

(A) inorganic impurities
(B) organic matter
(C) suspended solids
(D) micro-organisms

8. Cholera and typhoid in a locality can spread when:

(A) treated water is used to recharge groundwater
(B) biogas is burnt as a fuel
(C) sludge is converted into manure
(D) untreated sewage mixes with drinking water

9. Sewage is carried from the city to the place where it is cleaned. That place is the:

(A) sewage treatment plant
(B) septic tank
(C) overhead water tank
(D) river bank

10. The very first step when sewage enters a treatment plant is to pass it through:

(A) an aeration tank
(B) a sludge digester
(C) bar screens
(D) a chlorination chamber

11. Bar screens at the start of a treatment plant are meant to remove:

(A) dissolved nitrates
(B) harmful bacteria
(C) fine sand and grit
(D) large floating objects such as rags, cans and plastic bags

12. After the bar screens, the sewage is taken to a tank where its speed is reduced so that sand, grit and small pebbles can:

(A) settle to the bottom
(B) float to the surface as scum
(C) dissolve in the water
(D) turn into biogas

13. The speed of water is deliberately **slowed** in the grit-and-sand removal tank so that:

(A) bacteria can multiply faster
(B) oil rises more quickly
(C) the pipes do not burst
(D) heavy sand and pebbles get time to sink

14. In the settling tank, the solids that sink to the bottom are called:

(A) scum
(B) grit
(C) sludge
(D) biogas

15. In the settling tank, oil and grease that float to the top and are skimmed off are called:

(A) scum
(B) sludge
(C) manure
(D) silt

16. The settled sludge is removed from the bottom of the settling tank by:

(A) bubbling air through it
(B) burning it in place
(C) skimming it off the surface

(D) a scraper that pushes it to the centre/bottom

17. Which pair correctly matches the material with where it is collected in the settling tank?

(A) sludge — bottom; scum — top
(B) sludge — top; scum — bottom
(C) both at the top
(D) both at the bottom

18. The sludge collected from the settling tank is then sent to a:

(A) bar screen
(B) sundial
(C) digester
(D) grit chamber

19. Inside the sludge digester, the bacteria that act on the sludge work:

(A) in the absence of oxygen (anaerobically)
(B) only in bright sunlight
(C) in the presence of plenty of oxygen
(D) only at freezing temperatures

20. The anaerobic decomposition of sludge in the digester produces a useful gas called:

(A) oxygen
(B) biogas
(C) chlorine
(D) ozone

21. Biogas produced in the digester is valuable because it can be used as a:

(A) fertiliser sprayed on fields
(B) fuel
(C) disinfectant for drinking water
(D) building material

22. After digestion, the leftover digested sludge is dried and used as:

(A) drinking water
(B) bar-screen material
(C) scum
(D) manure

23. The clarified water leaving the settling tank is next sent to an aeration tank, where:

(A) sand is allowed to settle
(B) large objects are screened out
(C) the water is frozen
(D) air is bubbled through the water

24. Air is bubbled through the aeration tank mainly to:

(A) cool the water down
(B) help aerobic bacteria grow and consume the remaining organic waste
(C) push sand to the bottom
(D) produce biogas

25. The bacteria that clean up the leftover human waste, food and soaps in the aeration tank are:

(A) anaerobic bacteria
(B) not bacteria at all, but viruses
(C) aerobic bacteria
(D) the same bacteria used in the digester

26. The water that finally leaves a treatment plant is:

(A) always pure enough to drink straight away
(B) much cleaner, but not guaranteed to be safe drinking water
(C) dirtier than when it came in
(D) exactly the same as raw sewage

27. After treatment, the cleaned water is commonly:

(A) sent back to the bar screens (B) re-mixed with fresh sludge

(C) stored as biogas

(D) released into a water body or used to recharge groundwater

28. Good **sanitation** in a community most directly helps to:

(A) increase the number of waterborne diseases

(B) reduce the spread of diseases like cholera and typhoid

(C) speed up river pollution (D) raise the cost of biogas

29. Which of the following should **never** be thrown down the kitchen drain?

(A) plain water used to rinse a glass (B) a small amount of clean water

(C) water-soluble hand soap (D) used cooking oil

30. Throwing used cooking oil down the drain is discouraged mainly because it:

(A) makes the water taste sweet (B) speeds up the bar screens

(C) clogs pipes and hinders sewage treatment(D) is the best food for aerobic bacteria

31. A low-cost, on-site way to handle human waste for a single home where there is no sewer connection is a:

(A) septic tank (B) sewage treatment plant

(C) bar screen (D) grit chamber

32. Which of these is an environment-friendly toilet that turns human waste into manure using earthworms?

(A) a vermi-processing toilet (B) a flush-only toilet connected to a river

(C) an aeration tank (D) a sundial toilet

33. Put these treatment stages in the correct order: (i) aeration with air, (ii) bar screens, (iii) settling tank for sludge and scum, (iv) grit-and-sand removal.

(A) i → ii → iii → iv (B) ii → iv → iii → i

(C) iv → ii → i → iii (D) iii → i → ii → iv

34. Which step happens **earlier** in the plant?

(A) aeration of clarified water (B) drying of digested sludge into manure

(C) production of biogas (D) removal of grit and sand

35. A worker says, "Sand and pebbles are removed before the sludge settles out." This statement is:

(A) wrong — the settling tank comes first(B) wrong — sand is removed last of all

(C) correct — grit removal comes before the settling tank

(D) correct only on rainy days

36. Bar screens **cannot** remove which of the following from sewage?

(A) plastic bags (B) sticks and rags
(C) empty cans (D) dissolved nitrates and phosphates

37. Which statement about the aeration tank is correct?

(A) it relies on anaerobic bacteria (B) it removes large floating cans
(C) it relies on aerobic bacteria and a supply of air
(D) it produces biogas as its main job

38. The main reason a treatment plant must clean sewage before releasing it into a river is to:

(A) make the river water taste better
(B) prevent water pollution and the spread of disease
(C) increase the river's depth (D) provide grit for construction

39. Which sequence correctly traces a drop of dirty water from a home to a river?

(A) home → river → sewer → treatment plant
(B) treatment plant → home → sewer → river
(C) home → sewer → treatment plant → river
(D) home → treatment plant → sewer → river

40. A treatment plant's bar screen breaks down and lets plastic bags through. The most likely immediate problem is that:

(A) the next tanks and pipes get choked by the plastic
(B) biogas production doubles (C) the water instantly becomes drinkable
(D) aerobic bacteria die at once

41. Both the sludge digester and the aeration tank rely on bacteria. The key difference is that:

(A) the digester works without oxygen; the aeration tank needs oxygen
(B) the digester uses oxygen-loving bacteria; the aeration tank uses oxygen-hating bacteria
(C) neither uses bacteria (D) both need bright sunlight

42. By-products from the sludge digester that are actually **useful** are:

(A) scum and grit (B) plastic bags and rags
(C) biogas (fuel) and digested sludge (manure)(D) chlorine and ozone

43. After the bar screens and grit removal, the partly cleaned sewage goes next to the:

(A) river (B) settling tank
(C) home taps (D) overhead tank

44. "Treated water from a plant is safe to drink directly." This statement is:

(A) always true (B) true only in summer
(C) true because all bacteria are removed

(D) false — it is cleaner but usually still not fit for drinking

45. Untreated sewage discharged into a river is a major cause of:

(A) water pollution and waterborne diseases(B) cleaner drinking water downstream

(C) faster groundwater recharge (D) more biogas in homes

46. The contaminant in sewage that is **alive** and can directly cause illness is the:

(A) dissolved phosphate (B) disease-causing microbe

(C) suspended sand (D) floating cooking oil

47. Which of the following best explains why grit and sand are removed early, before the settling tank?

(A) so the heavy, abrasive particles don't damage equipment and don't clog the sludge handling later

(B) because sand is the most poisonous part of sewage

(C) because sand is needed to make biogas

(D) because the settling tank cannot hold any solids

48. Plant managers often run sewage slowly through tanks chiefly to give:

(A) the water time to evaporate (B) the pipes time to rust

(C) solids time to settle out by gravity (D) the gas time to escape into homes

49. Which practice helps keep public places sanitary and is usually managed by municipal bodies?

(A) dumping garbage into open drains

(B) regular cleaning of streets, drains and public toilets

(C) pouring chemicals down household sinks(D) blocking the sewers with plastic

50. A village builds a biogas plant fed with animal dung and human waste. The bacteria that produce its gas are:

(A) aerobic, needing lots of air (B) the same as those on bar screens

(C) anaerobic, working without oxygen (D) found only in the aeration tank

51. Which of these correctly describes scum?

(A) light, floatable matter such as oil and grease skimmed off the top

(B) heavy solids that sink to the bottom (C) the gas produced by the digester

(D) the manure made from dried sludge

52. If the aeration tank's air supply is switched off for a long time, the most likely result is:

(A) aerobic bacteria thrive even more (B) biogas is produced instead of clean water

(C) the aerobic bacteria struggle and organic waste is poorly cleaned

(D) the sludge turns into scum

53. Which of the following is the correct early-to-late order of what is **removed**?

(A) dissolved organic matter → grit/sand → big objects → scum

(B) big objects → grit/sand → settled sludge/floating scum → remaining dissolved organic matter

(C) settled sludge → big objects → grit → dissolved matter

(D) scum → dissolved matter → grit → big objects

54. A factory secretly pipes toxic chemicals straight into the sewer. The main danger is that these chemicals:

(A) can poison the helpful bacteria and pass into rivers, harming life

(B) make the bar screens work faster (C) turn instantly into harmless manure

(D) increase the drinking-water supply

55. The single biggest public-health benefit of building proper toilets and sewers in a town is:

(A) more electricity for homes

(B) fewer waterborne diseases such as cholera and dysentery

(C) taller buildings (D) cheaper plastic bags

56. Which of these is **not** an organic contaminant typically found in sewage?

(A) leftover food scraps (B) human faeces

(C) cooking oil (D) dissolved iron from a factory

57. Choose the correct statement about the order of biological treatment:

(A) both the sludge and the water are treated in the very same tank with no air

(B) sludge is digested anaerobically, while the separate clarified water is aerated aerobically

(C) the clarified water is digested without air and the sludge is aerated

(D) bacteria are used only on the bar screens

58. After all the treatment steps, what happens to the dried digested sludge?

(A) it is sent back to the bar screens (B) it is poured straight into the river untreated

(C) it is used as manure on fields (D) it is burnt to make scum

59. Why is it wise to avoid pouring medicines, tea leaves and chemicals down the drain?

(A) they help the aerobic bacteria grow (B) they make biogas burn brighter

(C) they turn sewage into drinking water

(D) they can choke pipes and harm the bacteria that treatment depends on

60. A student claims, "Once water passes the bar screen it has no impurities left." The claim is:

(A) correct — the bar screen removes everything

(B) correct only for rainwater (C) wrong — bar screens add new impurities

(D) wrong — bar screens only catch large objects; dissolved and microscopic impurities still remain

End of question paper. Maximum marks: 240. Marking: +4 correct, –1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 18: Wastewater Story

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	B	31	A	41	A	51	A
2	B	12	A	22	D	32	A	42	C	52	C
3	A	13	D	23	D	33	B	43	B	53	B
4	C	14	C	24	B	34	D	44	D	54	A
5	B	15	A	25	C	35	C	45	A	55	B
6	D	16	D	26	B	36	D	46	B	56	D
7	A	17	A	27	D	37	C	47	A	57	B
8	D	18	C	28	B	38	B	48	C	58	C
9	A	19	A	29	D	39	C	49	B	59	D
10	C	20	B	30	C	40	A	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Wastewater from homes, industries and farms is collectively called sewage.

2. (B) Sewage is a liquid that carries both suspended and dissolved impurities — it is not a solid, a gas, or clean rainwater.

3. (A) The system of underground pipes that carries sewage away is the sewerage.

4. (C) Washing water, kitchen-sink water and toilet flush are all household wastewater; molten iron from a furnace is not — it is industrial, not used water from a home.

5. (B) All the impurities in sewage are together called contaminants. (Nutrients and sediments are only part of the picture.)

6. (D) Faeces, food and oils come from living things, so they are organic matter (not inorganic salts or metals).

7. (A) Nitrates and phosphates are dissolved chemicals with no carbon-based living origin, so they are inorganic impurities.

8. (D) Waterborne diseases like cholera and typhoid spread when untreated sewage mixes with the drinking-water supply. Recharging groundwater with treated water, burning biogas and making manure do not spread them.

9. (A) Sewage is carried to a sewage treatment plant, where it is cleaned. (A septic tank serves a single building, not a city.)

10. (C) The first stage is the bar screens. Aeration and digestion come much later, and chlorination is not the opening step.

11. (D) Bar screens catch large floating objects — rags, cans, plastic bags. They cannot remove dissolved chemicals, bacteria or fine grit.

12. (A) Slowing the water lets the heavy sand, grit and pebbles settle to the bottom; they do not float, dissolve or become gas.

13. (D) Reducing the speed gives the heavy sand and pebbles time to sink under gravity. The slowing has nothing to do with bacteria, oil or pipe pressure.

14. (C) Solids that sink to the bottom of the settling tank are called sludge.

15. (A) Light floatables such as oil and grease that are skimmed off the top are called scum.

16. (D) Settled sludge is removed by a scraper that moves it along the tank floor — not by aeration, burning or surface skimming.

17. (A) Sludge is heavy and settles at the bottom; scum is light and floats at the top. So: sludge — bottom, scum — top.

18. (C) The collected sludge is sent to a digester, where bacteria act on it.

19. (A) In the digester the bacteria work without oxygen — anaerobically. Sunlight, abundant oxygen and freezing are not involved.

20. (B) Anaerobic digestion of sludge releases biogas, a useful fuel — not oxygen, chlorine or ozone.

21. (B) Biogas can be burnt, so it is valued as a fuel — not as a fertiliser, disinfectant or building material.

22. (D) After digestion the dried sludge is used as manure. It is certainly not drinking water or scum.

23. (D) The clarified water is sent to an aeration tank, where air is bubbled through it. (Settling and screening happen earlier.)

24. (B) Bubbling air supplies oxygen so aerobic bacteria can grow and eat the remaining organic waste — not to cool the water or make biogas.

25. (C) The aeration tank relies on aerobic bacteria (they need the bubbled air). Anaerobic bacteria belong to the digester.

26. (B) The outgoing water is much cleaner but is not guaranteed safe to drink — a classic trap. It is certainly not dirtier or unchanged.

27. (D) Cleaned water is released into a water body or used to recharge groundwater. It is not returned to the bar screens or stored as gas.

28. (B) Good sanitation reduces the spread of diseases like cholera and typhoid; it does not increase disease or pollution.

29. (D) Used cooking oil must never go down the drain — it clogs pipes and hampers treatment. Plain or soapy rinse water is fine.

30. (C) Used cooking oil clogs pipes and disrupts treatment; it does not sweeten water, speed up screens, or feed the bacteria usefully.

31. (A) A septic tank is the low-cost, on-site option for a single home with no sewer connection. A full treatment plant, bar screen or grit chamber are parts of a municipal system.

32. (A) A vermi-processing toilet uses earthworms to turn human waste into manure.

33. (B) The correct order is bar screens (ii) → grit-and-sand removal (iv) → settling tank (iii) → aeration (i).

34. (D) Grit and sand are removed near the start, long before aeration, sludge drying or biogas production. So grit-and-sand removal happens earlier.

35. (C) Grit removal does come before the settling tank, so the worker's statement is correct.

36. (D) Bar screens catch only large solids; they cannot remove dissolved nitrates and phosphates. They do catch plastic, sticks, rags and cans.

37. (C) The aeration tank relies on aerobic bacteria plus a supply of air. It is not anaerobic, does not screen cans, and does not mainly make biogas.

38. (B) Sewage is treated before release to prevent water pollution and disease — not to change the river's taste, depth or supply grit.

39. (C) A drop travels home → sewer → treatment plant → river. The other orders put the river or plant in the wrong place.

40. (A) If plastic bags get past the broken bar screen, they choke the following tanks and pipes. They do not boost biogas or purify water.

41. (A) The digester works without oxygen (anaerobic); the aeration tank needs oxygen (aerobic). That is the key difference between the two bacterial stages.

42. (C) The digester's useful by-products are biogas (a fuel) and digested sludge (manure). Scum, grit, plastic and chlorine are not its products.

43. (B) After bar screens and grit removal, the partly cleaned sewage flows to the settling tank — not straight to the river or the taps.

44. (D) Treated water is cleaner but usually still not fit to drink directly — so the claim is false. (This is the standard trap.)

45. (A) Untreated sewage in a river causes water pollution and waterborne diseases — not cleaner water or more biogas.

46. (B) The living contaminant is the disease-causing microbe. Phosphate, sand and oil are not alive.

47. (A) Grit and sand are abrasive and heavy; removing them early protects equipment and prevents them clogging the sludge handling later. They are not the most poisonous part and are not needed for biogas.

48. (C) Running the sewage slowly lets solids settle out by gravity — the whole point of the settling and grit tanks. It is not about evaporation, rust or gas.

49. (B) Municipal bodies keep public places sanitary by regularly cleaning streets, drains and public toilets — not by dumping or blocking.

50. (C) A biogas plant's gas comes from anaerobic bacteria working without oxygen, just like the digester. Aerobic bacteria need air and would not make the gas.

51. (A) Scum is the light, floatable matter (oil and grease) skimmed off the top — not the heavy sinking sludge, the gas, or the manure.

52. (C) Without bubbled air the aerobic bacteria struggle, so the organic waste is poorly cleaned. They do not thrive, and the tank does not start making biogas.

53. (B) What is removed runs early-to-late: big objects (bar screens) → grit/sand → settled sludge and floating scum → remaining dissolved organic matter (aeration).

54. (A) Toxic chemicals can poison the helpful bacteria and pass on into rivers, harming life. They do not speed up screens or become manure.

55. (B) Proper toilets and sewers chiefly bring fewer waterborne diseases such as cholera and dysentery — not electricity, taller buildings or cheaper plastic.

56. (D) Food scraps, faeces and cooking oil are organic; dissolved iron from a factory is inorganic, so it is the odd one out.

57. (B) The sludge is digested anaerobically (no air) while the separate clarified water is aerated aerobically (with air). The two biological stages are distinct.

58. (C) The dried digested sludge is used as manure on fields — not returned to screens, poured into rivers, or burnt.

59. (D) Medicines, tea leaves and chemicals can choke pipes and harm the bacteria that treatment relies on, so they should be kept out of drains.

60. (D) Bar screens only catch large objects; dissolved and microscopic impurities remain. So the claim that no impurities are left is wrong.

Solutions complete: 60 / 60.